Srihari Bandarupalli

Hyderabad, Telangana
+91-6302481566 | sriharib128@gmail.com | LinkedIn | sriharib128

Education

International Institute of Information and Technology, Hyderabad — **Nov. 2021 – May 2026**
B.Tech with MS by Research in ECE — ***CGPA: 9.2*** · *Department Gold Medal*

Experience

BharatGen — **Sept 2025 – Present**
Research Engineer

- First-authored **DocLayout-VL** *(accepted to ECCV'26)*, a foundation model reframing document layout analysis as open-set, hierarchical, **promptable segmentation**—a VLM generates the layout tree and a novel hierarchy-aware grounding module produces pixel-accurate masks, **outperforming frontier VLMs (Gemini 3 Pro, GPT-5.2) and per-dataset specialist detectors zero-shot across 10 public benchmarks**.
- Built the data programme for **Patram 1.5**, BharatGen's document foundation model: designed a task taxonomy of **186 tasks across 12 capabilities** with a machine-checked compatibility matrix, generated an **8B-token multimodal corpus** via large-scale teacher distillation, and am **now training a 7B model on 128 GPUs**.
- Engineered the team's **custom distributed trainer** built on top of Hugging Face and DeepSpeed ZeRO-3: pinned non-compute tensors to CPU and reduced validation metrics per-rank before any cross-GPU gather, eliminating OOMs and raising per-GPU batch size 1 → 6–8; Tuned **multi-node communication for AWS EFA**, retaining **97.8%** of all-reduce bandwidth across nodes.
- Built a **crash-safe, exactly-once distributed data engine** for 100M+ image GPU-inference workloads, with **DAG pipelines** supporting fan-out, fan-in, and **per-item routing decided at runtime**—scaled one job unattended from 5 to **24 nodes (192 GPUs)** at 99.98% completion, and lifted throughput **29%** on identical hardware by eliminating GPU idling via shared-memory IPC.
- Wrote **Patram-FS**, a per-node Go storage daemon serving multi-TB datasets from S3 through local NVMe—cutting storage requests per training run from ∼10M to ∼200 (**50,000**×), speeding annotation reads **11**× via kernel page-cache warming, and taking the shared filesystem off the training data path; adopted across the team's training and data pipelines.

Sprinklr — **May 2025 – July 2025**
Machine Learning Intern

- Built an enterprise **VLM evaluation framework** on production-like data distributions *(published, AI-ML Systems'25)*, and showed that targeted data curation lets a fine-tuned **Qwen-7B match Qwen-32B accuracy at 80% lower compute**, consolidating 10+ task-specific visual systems into one model.

Language Technologies Research Centre (LTRC), IIIT-H — **May 2023 – Present**
Undergraduate Researcher, Speech Processing Lab — *Prof. Anil Kumar Vuppala*

- Demonstrated that phonetic similarity–guided cross-lingual data selection for **low-resource ASR** can substitute for model scaling, matching larger models with significantly fewer parameters. Two publications (Interspeech'25, IJCNLP-AACL'25).

Publications

- *DocLayout-VL: A Foundational Model for Hierarchical, Open-Set and Promptable Document Layout Segmentation.* **(ECCV'26)**
- *Efficient ASR for Low-Resource Languages: Leveraging Cross-Lingual Unlabeled Data.* (**IJCNLP-AACL'25**)
- *VLMs-in-the-Wild: Bridging the Gap Between Academic Benchmarks and Enterprise Reality.* (**AI-ML Systems'25**)
- *End-to-End Speech Translation for Low-Resource Languages Using Weakly Labeled Data.*(**Interspeech'25**)

Projects

Strife—Distributed Payment Gateway | *Go, gRPC* — **Link**

- Implemented 2-Phase Commit across bank servers with mutual TLS, idempotent transactions, offline payment queuing, and WAL-based crash recovery.

Distributed File System | *C++, MPI* — **Link**

- Implemented fault tolerance via heartbeat monitoring, replication, load-balanced distribution, and distributed search.

Technical Skills

Languages: Python, C/C++, Go, Bash, SQL
ML Systems: PyTorch, DeepSpeed ZeRO, HuggingFace, vLLM, Fairseq, Kaldi
Infrastructure: PostgreSQL, Kubernetes, Docker, AWS (S3, EFA), SLURM, Prometheus/Grafana, gRPC, MPI

Achievements

- **Department Gold Medal**, IIIT Hyderabad, for exceptional academic achievement.
- **Dean's List** for academic excellence, four consecutive years (2021–2025).